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Physics-informed neural network for predicting orbital parameters of Low Earth Orbit satellites using two line element dataset ☆

Md Sahat Mahmud, Yiping Jiang *, Ruirui Liu, Zihong Zhou, Bing Xu

Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, 999077, the Hong Kong Special Administrative Region of China

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Abstract

This research introduces a novel Physics-Informed Neural Network (PINN) framework using Long Short-Term Memory (LSTM) architecture as the baseline model for the precise prediction of orbital parameters of Low Earth Orbit (LEO) satellites, relying exclusively on publicly available Two-Line Element (TLE) datasets. The proposed method is validated for near-circular orbit with a near zero eccentricity in LEO orbits, which is a characteristic of operational LEO constellations such as Iridium, Orbcomm, Starlink, etc. The main goal is to address the challenges linked to infrequent updates of TLE data, which generally decreases propagation accuracy with time. The proposed method efficiently incorporates fundamental physical laws of orbital mechanics into the neural network's learning framework, improving prediction accuracy, stability, and generalizability relative to traditional solely data-driven approaches. Results from the experiments conducted utilizing historical TLE records of the IRIDIUM 7 and ORBCOMM FM26 satellites reveal that the integration of physics-based constraints significantly reduces prediction errors for satellite position and velocity. For IRIDIUM 7 the position and velocity estimates are improved by 87% and 70% respectively, which are 50% and 72% respectively for ORBCOMM FM26 satellite. The proposed PINN also demonstrates notable improvement over the baseline LSTM, with most R^2 improvement for inclination and mean anomaly predictions by approximately 59.5% and 59.1%, respectively. These improvements are crucial for high-precision applications such as Positioning, Navigation, and Timing (PNT) as well as space traffic management in the increasingly crowded LEO environment, where high fidelity satellite ephemerides are important.
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Keywords: Physics informed neural networks; Two line element; Orbital parameter prediction

1. Introduction

In recent years, the rapid growth of active satellite constellations and the accumulation of orbital debris, particularly in Low Earth Orbit (LEO), have raised concerns regarding the long-term sustainability of Earth's orbits (Martinez, 2023). With the current pace of expansion, the risks of in-orbit collisions and debris proliferation are drastically elevated. However, the use of outdated or inaccurate orbital data for these space objects can lead to missed or false collision warnings. One prominent example of such issue is the reliance on publicly available Two Line Element (TLE) files, which remain a primary source of orbital information for space situational awareness, but are known to suffer from significant biases and errors (Bizalion et al., 2023). To address this challenge, the development of high-reliability orbit determination for conjunction assessments is an essential prerequisite for effective space traffic management and sustainable operations (Virgili et al., 2016).

At the same time, LEO mega-constellations such as SpaceX's Starlink and Iridium NEXT have received significant interest due to their potential in providing opportunistic Positioning, Navigation, and Timing (PNT) services

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* Corresponding author.
E-mail address: yiping.jiang@polyu.edu.hk (Y. Jiang).

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(Haidar-Ahmad et al., 2022). With thousands of LEO satellites now planned or deployed, the dense coverage, strong signals these networks offer can be used to augment the traditional Global Navigation Satellite Systems (GNSS) for navigation services. Researchers have begun exploiting Starlink communication signals for opportunistic PNT, demonstrating meter-level positioning accuracy using LEO satellites as an alternative to GNSS (Haidar-Ahmad et al., 2022). Hence, accurate orbit knowledge is now crucial for both space-sustainability initiatives and advanced LEO based PNT capabilities, making improved prediction techniques a shared and urgent priority for the space industry.

Tracking LEO satellites presents a fundamental difficulty given the community's dependence on TLE data for orbital parameters. Published by organizations such as North American Aerospace Defense Command (NORAD), TLE sets give each satellite's publicly accessible mean orbital components. These components then can be used for orbit propagation with the help of dynamic models like Simplified General Perturbations 4 (SGP4). While convenient and widely used, TLEs suffer for limited update frequency, which eventually degrades the orbit propagation accuracy over time. In practice, TLE has an update rate of 1.7 per day (Chen and Lin, 2023) in average, meaning a satellite's published orbit can be many hours old. Using SGP4 to extrapolate the orbit leads to increasing errors between updates due to unmodelled forces or maneuvers. One study reported an initial ~2.5 km orbit error at the initial epoch increased to over 100 km after only one week of propagation without new data (Kovar et al., 2022). Similarly, initial orbit solution for LEO satellites can be off by a few kilometers, with errors compounding until the next TLE update (Haidar-Ahmad et al., 2022). This level of inaccuracy is not acceptable for precision and sensitive applications such as LEO PNT and space traffic management.

To address the constraints of sparse and coarse TLE data, numerous orbit determination and prediction methodologies have been explored, encompassing conventional numerical propagators that incorporate intricate perturbations e.g., Earth's unevenly distributed gravitational field, solar radiation pressure and filtering techniques e.g., Kalman Filter, Particle Filter, Batch Least Squares Estimation (Shorten et al., 2025; Levit and Marshall, 2011). Although physics-based models offer high precision, they often depend on accurate and regularly updated orbital parameters, which are generally inaccessible for commercial constellations such as Starlink, Iridium, and Orbcomm. This limits the practical deployment of such models for these satellites (Sun et al., 2025). Alternatively, machine learning methodologies have been investigated. Long-Short Term Memory (LSTM), a type of Recurrent Neural Network (RNN), was used to forecast orbital elements based on prior TLE data (Mahmud et al., 2024). Moreover, hybrid approaches that integrate physical modelling such as Extended Kalman Filters with neural networks demonstrated a 90% reduction in ephemeris errors relative to independent SGP4 propagation (Haidar-Ahmad et al., 2022). Notwithstanding these achievements, solely data-driven approaches necessitate substantial training data and encounter difficulties in generalizing under fluctuating orbital contexts. Recent investigations on amplitude scintillation detection (Li et al., 2024) and equatorial plasma bubble modeling (Li and Jiang, 2024) further emphasize the benefits of incorporating physics-based insights into data-driven models.

Recently, Physics-Informed Neural Networks (PINNs) have arisen as a formidable alternative, incorporating established physics directly into the learning framework. PINNs utilize governing orbital equations to limit their predictions, necessitating fewer data samples, maintaining physical validity, and demonstrating enhanced robustness, even under conditions of inadequate historical data or unclear orbital parameters (Varey et al., 2024). However, existing PINN solutions employ static Feed-Forward Neural Networks (FFNN) (Scorsoglio et al., 2023), which do not inherently capture the sequential and temporal correlations within historical orbital data. This limits the overall prediction accuracy of the existing models limited over extended time horizons. Additionally, existing PINN approaches often assume availability of high-fidelity and regularly sampled observational data (Scorsoglio et al., 2023; Martin and Schaub, 2022), which is scarce for many LEO satellites, especially newer ones.

In this work, we propose a PINN based framework, with LSTM as the backbone, for improved orbit prediction of LEO satellites focusing on the prediction of the six orbital elements defined by the TLE records: inclination (i), right ascension of the ascending node (Ω, RAAN), eccentricity (e), argument of perigee (ω), mean anomaly (M), and mean motion (n). The key novelty is the integration of LSTM and regularizing the PINN with a customized loss function, composed of data loss and physics loss, which effectively instructs the neural network to obey the physics laws. Additionally, the usage of TLE records as training data is advantageous due to their easy accessibility and wide coverage. The physics loss is estimated by several governing equations, which are differential equations related to the variation of each element. This allows the network to leverage the knowledge of orbital mechanics (Keplerian motion, perturbation effects, etc.) to fill in the gaps between sparse TLE updates. The model was assessed on a comparable time span to the typical TLE update intervals and was compared against the baseline LSTM model trained with the same data. The outputs were then fed into the standard SGP4 propagator to generate the satellite state at specific future epochs for performance evaluation. Ultimately, this research introduces a physics-informed predictive modeling approach that substantially enhances orbital parameter estimation accuracy, thereby directly improving conjunction assessment reliability, orbital debris mitigation effectiveness, and precision of LEO PNT services. By explicitly incorporating physical constraints

into deep learning frameworks, this methodology facilitates superior orbital state predictions, advancing operational integrity and contributing to the overarching objective of sustainable and secure space operations.

The remainder of this paper is as follows: Section 2 introduces the neural network used in this work. Section 3 details the governing equations used, proposed model architecture, data pre-processing and training process. Section 4 presents experimental results, demonstrating the generalization capabilities and effectiveness of the current model in satellite position and velocity estimation. Finally, Section 5 discusses the conclusion and future works, focusing on potential improvements of the model and application for LEO PNT and space sustainability.

2. Neural networks

2.1. Long short-term memory

Although RNNs perform well while capturing temporal dependencies in historical data, they often encounter challenges in modeling long-term dependencies due to the vanishing gradient problem. This issue arises during training when gradients diminish significantly, restricting the network's capability to update weights based on distant input information effectively (Hochreiter et al., 2001). To address this problem, LSTMs introduce a special gating mechanism which allows them to regulate the flow of information through the network (Hochreiter and Schmidhuber, 1997). The gates are called – input gate, output gate and forget gate – among which the forget gate allows the network to discard irrelevant past information, input gate updates the cell state with new relevant data, and the output gate controls what information is passed forward in the network. This architecture allows LSTM to retain and manipulate temporal dependencies over extended sequences, making them effective particularly for time series forecasting. Fig. 1 (Mahmud et al., 2024; Gers and Schmidhuber, 2000) illustrates the architecture of LSTM networks with the forget gate.

The basic elements of LSTM network are described by the cell states and their controlling gates. The cell states serve as the network's memory unit by preserving essential information during sequence processing. Control gates, which act as regulators, are implemented using analog operations that use element-wise multiplication with sigmoid and hyperbolic tangent (tanh) functions (Osama et al., 2022). The sigmoid function determines the importance of an input by scaling it between a value from 0 to 1, thereby selectively managing the information flow through the network. The tanh function scales the inputs within a range from −1 to 1, which controls the gradient magnitudes to prevent excessively large values. Each gate obtains its specific weight during training through neural network learning mechanisms. The gates learn to determine the importance of incoming information by distinguishing between essential and irrelevant inputs. The mathematical description of the network can be formulated through the following equations based on the structural representation shown in Fig. 1 (Gers and Schmidhuber, 2000):

$$\sigma(x) = \frac{1}{1+e^{-x}} \tag{1}$$

$$\tanh(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}} \tag{2}$$

$$f_t = \sigma\left(W_f h_{t-1} + W_f x_t + b_f\right) \tag{3}$$

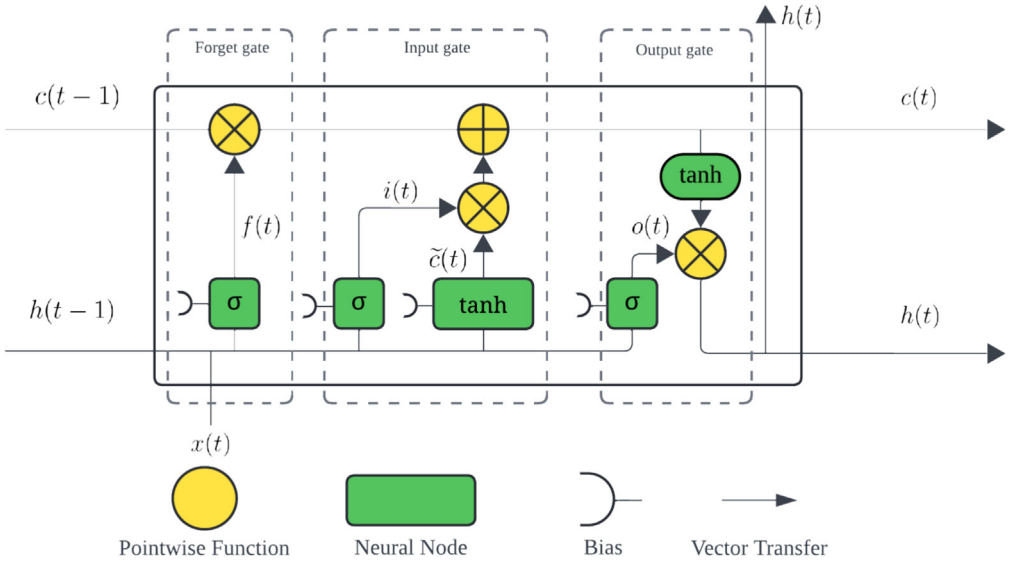


Fig. 1. This figure illustrates an LSTM cell's structure and data flow. The Forget Gate (left) computes f(t) to determine what information to discard from the previous cell state c(t-1). The Input Gate (middle) decides what new information to add to the cell state by computing it and generating candidate values $\tilde{c}(t)$, updating the cell state to c(t). The Output Gate (right) determines the output hidden state by computing o(t) and producing h(t) (Mahmud et al., 2024).

$$i_t = \sigma(W_i h_{t-1} + W_i x_t + b_i) \tag{4}$$

$$\tilde{c_t} = \tanh\left(W_c \tilde{h}_{t-1} + W_c \tilde{x_t} + b_c\right) \tag{5}$$

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c_t} \tag{6}$$

$$o_t = \sigma(W_o h_{t-1} + W_o x_t + b_o) \tag{7}$$

$$h_t = o_t \cdot \tanh(c_t) \tag{8}$$

f_t here is the forget gate that determines which information from the previous cell state c_{t-1} should be discarded, depending on the value which ranges from 0 to 1, 0 meaning complete forgetting while 1 represents complete retention. This value is determined by the sigmoid activation function σ. Based on the latest information, the input gate i_t updates the values of cell state. This task is also controlled by the sigmoid activation function similar to the forget gate. Determined by the hyperbolic tangent activation function, the candidate value for updating the cell state $\tilde{c_t}$ is scaled between -1 to 1. The update mechanism of cell state c_t, is shown by Eq. (6), which allows the LSTM network to effectively maintain important information over extended sequences. The output from the network is modulated by the output gate o_t, which uses σ to scale the outputs between 0 and 1. Finally, the hidden state h_t, utilized in subsequent steps, is derived by applying the tanh function to the updated $\tilde{c_t}$, and then multiplying this result elementwise by o_t.

However, raw LSTM networks focus purely on mapping the inputs to outputs based on historical data, without incorporating physical constraints into the network or its loss function (Hochreiter and Schmidhuber, 1997). This may result in inaccurate predictions violating the physics laws when exposed to new data or conditions. To address this problem, Physics-Informed Neural Networks (PINNs) were proposed to incorporate fundamental physical principles into the training process directly, ensuring that the predicted results remain consistent with the established domain knowledge throughout (Raissi et al., 2019).

2.2. Physics-informed neural network

PINNs are a class of deep learning models that integrate physical laws directly into the neural network's learning process (Cuomo et al., 2022). Physical laws are often expressed as differential equations. The differential governing equations serve as constraints or regularizers during training, which inherently forces the networks predictions to obey the known physics. The method combines machine learning with traditional computational modeling by including domain knowledge (such as Newton's laws or Navier-Stokes equations) into the model itself (Loshelder et al., 2025). This approach alleviates the concerns of deep learning models being treated as 'Black box', and provides more trustworthy and transparent solutions for sensitive systems such as autonomous driving, satellite navigation, etc.

PINNs utilize neural networks' universal approximation properties to solve physical system equations (Raissi et al., 2019). The task of solving differential equations numerically transforms into an optimization problem using PINNs, which avoids traditional numerical solution methods (Cuomo et al., 2022). A neural network $u_\theta(x)$, with trainable parameters θ, approximates state variables through input coordinate functions x, which can contain spatial and temporal information. The loss function $\mathscr{L}(\theta)$ used for network training receives additional physics-based terms as part of its composition. $\mathscr{L}(\theta)$ consists of two parts which include both data mismatch and terms that enforce differential equation residuals and boundary and initial conditions. The network receives penalties during training based on its failure to generate outputs which follow the governing physical laws. The total loss function used for training the PINN is then defined as a weighted sum of multiple components:

$$\mathscr{L}(\theta) = \omega_F \mathscr{L}_{\mathscr{F}}(\theta) + \omega_D \mathscr{L}_{\mathscr{D}}(\theta) \tag{9}$$

$$\theta^* = \arg\min_{\theta} \mathscr{L}(\theta) \tag{10}$$

where $\mathscr{L}_{\mathscr{F}}$ is the physics loss, computed as the mean squared residual of the Differential Equation (DE, an equation that relates a function to its derivatives, describing how a system evolves over time or space):

$$\mathscr{L}_{\mathscr{F}}(\theta) = \frac{1}{N_c} \sum_{i=1}^{N_c} \left|\mathscr{F}[u_\theta](x_i)\right|^2 \tag{11}$$

with $\{x_i\}_{i=1}^{N_c}$ being the collocation points in the domain. $\mathscr{L}_{\mathscr{D}}$ is the data loss, which can be defined as:

$$\mathscr{L}_{\mathscr{D}}(\theta) = \frac{1}{N_d} \sum_{j=1}^{N_d} \left|u_\theta(z_j) - u_j^{(\text{obs})}\right|^2 \tag{12}$$

z_j are the locations of the data points observed with values $u_j^{(}$obs). $\omega_F, \omega_B, \omega_D$ are the weight coefficients defined by the used to balance the different loss terms during training. By minimizing $\mathscr{L}(\theta)$, the network learns a function $u_\theta(x)$ that simultaneously fits available data and satisfies physical laws as closely as possible, thereby converting the problem of solving differential equations into a machine learning optimization problem. The schematic architecture of a PINN is shown in Fig. 2.

According to the architecture, a PINN can be considered as a form of multi-task learning in which the network is simultaneously trying to regress to given data and satisfy the different equations. In cases where no direct observations are available, a PINN can operate in a completely unsupervised manner by relying solely on the DE and boundary conditions for training the model (Cuomo et al., 2022). This significantly differs from conventional deep learning, which generally requires a significant quantity of labelled data. PINNs employ automatic differentiation (AD) to calculate the required derivatives of $u_\theta(x)$, allowing the uncomplicated estimation of the DE residuals

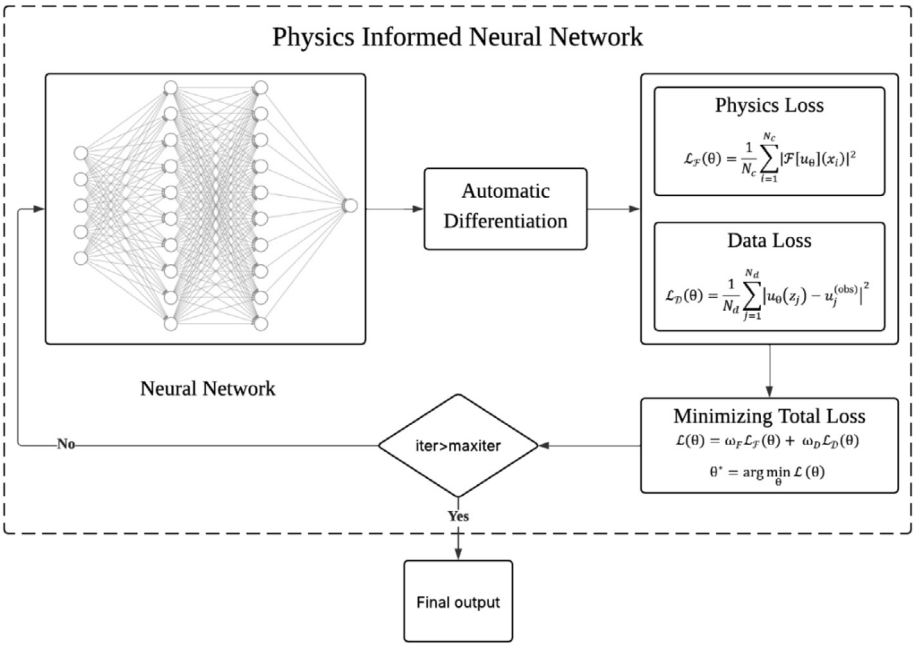


Fig. 2. Flowchart illustrating the training procedure of PINNs. The neural network outputs are differentiated automatically to evaluate the physics-based loss $\mathscr{L}_{\mathscr{F}}$ enforcing the governing equations and the data-based loss $\mathscr{L}_{\mathscr{D}}$ enforcing data fidelity. Both losses are combined with weights $\omega_{\mathscr{F}}$ and $\omega_{\mathscr{D}}$ to form the total loss, which is iteratively minimized until the stopping criterion (maximum iterations) is reached.

inside the loss function (Raissi et al., 2019). AD enables the exact implementation of differential operators on neural network outputs, therefore encoding the underlying physics without the need of deriving explicit finite difference formulas.

3. PINN governing equations for orbital parameter prediction

The loss functions of each of the models are modified with the respective governing equations of the orbital elements. For each of the elements, relevant differential equations-the Keplerian dynamics or Gauss's Variational equations derived from perturbation theory-are incorporated into the loss to ensure the models consistently obey the physics laws evolving with perturbative forces and physical orbital motion. For most of the models, a simpler approximation of atmospheric drag (Eq. (17)) and J_2 effect was considered as in the LEO regime, the orbital decay due to drag and the perturbative effects of Earth's J_2 are the dominant non-Keplerian influences. For near-circular orbits ($e \ll 1$, typically $e < 0.01$ for most operational LEO constellations), the periodic and other secular variations are relatively small and can be well approximated by the averaged forms used here. The simplified approach captures the essential physical behavior without requiring a full high-fidelity force model, without overcomplicating the network.

Since TLE data already provides a first derivative of mean motion, n, the proposed PINN model uses the measurements as a baseline to compute the Mean Absolute Error (MAE) with the derivative of the network's predicted n in which this approach decreases the computational load on the overall system and increases the accuracy of the baseline. The general Gauss variational equation for the change of i is (Montenbruck and Gill, 2000):

$$\frac{di}{dt} = \frac{r\, a_N \cos(f+\omega)}{h} \tag{13}$$

where r is the instantaneous orbital radius, a_N is the normal component of the perturbing acceleration, f is the true anomaly, ω is the argument of perigee, and h is the specific angular momentum defined by $h = \sqrt{\mu\, a\,(1-e^2)}$ with μ being the earth's gravitation parameter.

For the purpose of deriving the J_2 effect on orbital inclination, the normal component $a_N^{J_2}$ can be expressed as (Montenbruck and Gill, 2000):

$$a_N^{J_2} = -\frac{3J_2\,\mu\,R_e^2}{2r^5}\sin i\cos i\sin(f+\omega) \tag{14}$$

Here, R_e is the radius of the earth, J_2 is the earth's second zonal harmonic. Substituting $a_N^{J_2}$ in Eq. (13), we have:

$$\frac{di}{dt} = \frac{r\cos(f+\omega)}{h}\left[-\frac{3J_2\,\mu\,R_e^2}{2r^5}\sin i\cos i\sin(f+\omega)\right] \tag{15}$$

Applying trigonometric identity, this simplifies to:

$$\frac{di}{dt} = -\frac{3J_2\,\mu\,R_e^2}{4h\,r^4}\sin i\cos i\sin[2(f+\omega)] \tag{16}$$

For near-circular orbits, where $e \ll 1$, the short period oscillatory term $\sin[2(f+\omega)]$ is often replaced by an effective constant amplitude. With the approximation $r \approx a$, the

specific angular momentum $h \approx \sqrt{\mu a}$ and the drag correction with a tunable parameter K_{drag}, a simplified decay model is obtained:

$$\Delta a = -K_{\text{drag}} B^* \Delta t \tag{17}$$

The atmospheric drag is treated as a perturbation affecting primarily a, with secondary effects on other elements captured implicitly through data learning. This simplification is appropriate as: (1) PINN is fundamentally data-augmented physics, not physics only modelling. In other words, the network learns empirical TLE evolution while physics prevents unrealistic extrapolation and (2) the dual loss formulation allows the data component to compensate for physics simplifications when needed.

Hence, by incorporating the general Gauss variational equations with J_2 perturbation to avoid the need for direct velocity measurements, the governing equation used for the PINN model of i is derived to be

$$\frac{di}{dt} = \frac{3}{4} J_2 \left(\frac{R_e}{a_{\text{eff}}}\right)^2 n \sin i \cos i \tag{18}$$

where $a_{\text{eff}} = a + \Delta a$, is the drag-corrected a and it is used in all subsequent governing equations to account for atmospheric drag effects in a simplified manner.

For the model of M, the governing equation can be expressed as (Montenbruck and Gill, 2000):

$$\frac{dM}{dt} = n + \frac{3}{4} J_2 \left(\frac{R_e}{a_{\text{eff}}}\right)^2 n\left(3\cos^2 i - 1\right) \tag{19}$$

For Ω and ω, the following governing equations were used respectively with drag correction to a (Montenbruck and Gill, 2000):

$$\frac{d\Omega}{dt} = -\frac{3}{2} J_2 n \left(\frac{R_e}{a_{\text{eff}}(1-e^2)}\right)^2 \cos i \tag{20}$$

$$\frac{d\omega}{dt} = \frac{3}{4} J_2 n \left(\frac{R_e}{a_{\text{eff}}(1-e^2)}\right)^2 \left(5\cos^2 i - 1\right) \tag{21}$$

For the satellites in LEO, especially those with a near-circular orbit, atmospheric drag is the only dominant cause to decay in e. Additionally, as TLE only provides averaged orbital elements not the precise orbital states of a satellite, it is difficult to compute the full Gauss variational equation for e. As e has a low variation in the LEO regime over time, the model can easily capture the trend in e even with solely data driven approaches (Mahmud et al., 2024). However, a simple empirical governing equation has been used to softly account for the e decay, which can be written as:

$$\frac{de}{dt} \approx -\gamma e \tag{22}$$

where γ is a tunable parameter, the best value of which was found through experiments on the model. For a PINN, where the goal is to impose physical consistency, a theoretically efficient model is preferable. Eq. (22) presents a soft constraint that is simple to use and adjust, therefore regularizing the learning process without complexing the loss function.

4. Orbital parameter prediction with proposed PINN

4.1. System architecture

The proposed system architecture is designed to learn the temporal evolution of TLE parameters using a PINN, constructed on top of a LSTM network. Six separate PINN models are proposed, each tailored for estimating one of the six orbital elements: n, i, e, Ω, ω, and M. a was not modeled explicitly, as it can be deterministically derived from n through Kepler's third law. Excluding it as a separate prediction target reduces redundancy and helps maintain a lightweight model architecture without compromising completeness. Each PINN model is designed independently but operates sequentially to refine predictions iteratively, resulting in a comprehensive and physically consistent orbital parameter prediction. The following subsections will discuss the data preprocessing methods and model training approaches used in the proposed PINN model.

Although a fully independent parallel architecture might appear simpler, the sequential approach is adopted because orbital elements exhibit strong physical interdependencies (e.g., a is derived from n, both defining orbit size). Sequential prediction leverages these relationships by using earlier predictions as inputs to later models, improving accuracy. Additionally, this approach naturally accommodates progressive feature availability – early models use only basic features (T, D_s, D_m), while later models benefit from previously predicted elements.

4.1.1. Data pre-processing

Prior to training the model, raw TLE data needs to go through a preprocessing pipeline designed to enhance data suitability for training purposes. Raw TLE consists of outliers, spikes and fluctuations, which may degrade the performance of the model. The preprocessing pipeline first extracts the relevant orbital parameters and filters out the outliers. Other preprocessing steps include:

(1) Converting the raw Ω to sin Ω as sudden fluctuations may cause discontinuity,
(2) Converting the i to sin i,
(3) Calculating and appending the values of Earth-to-Sun distance (D_s) and Earth-to-Moon distance (D_m) in the TLE data frame,
(4) Calculating and appending semi-major axis (a) and true anomaly (v) into the data frame using the basic orbital mechanics formulas shows in Eqs. (23) and (25) below (Kidder and Haar, 1995),
(5) Normalizing the training dataset since the orbital elements are all measured in different scales.

$$a = \left(\frac{\mu}{\left(\frac{2\pi n}{86400}\right)^2}\right)^{\frac{1}{3}} \quad (23)$$

$$M = E - e\sin(E) \quad (24)$$

$$\tan\left(\frac{v}{2}\right) = \sqrt{\frac{1+e}{1-e}}\tan\left(\frac{E}{2}\right) \quad (25)$$

where a, n, M, E, e and v are semi-major axis, mean motion, mean anomaly, eccentric anomaly and true anomaly respectively.

The auxiliary input features D_s and D_m are deterministic input features derived from JPL ephemerides, not learned parameters or network outputs. They are computed for each TLE epoch to provide the model with additional context about gravitational perturbations affecting orbital dynamics. These distances are not modified during the training process and are not used in the satellite state propagation stage. Their inclusion as input features was motivated by empirical results from prior work (Mahmud et al., 2024) which demonstrated significant performance improvements when additional features were introduced to the LSTM model.

Min-max normalization is applied to map all values to z within the range of 0–1, where z is calculated as $\frac{x-\min(x)}{\max(x)-\min(x)}$. The distances from Earth to the Sun and from Earth to the Moon are calculated for each TLE epoch utilizing Python's current libraries. These libraries utilize ephemeris data from the Jet Propulsion Laboratory (JPL) to compute high-precision ephemerides. Uniformly sampled time sequences are used to impose temporal alignment, hence supporting LSTM training's sequential nature. Fixed-length time frames allow historical TLE data to be broken out into input sequences reflecting the temporal evolution of orbital elements over multiple epochs.

4.1.2. Model training

Upon completion of data pre-processing, the subsequent orbital elements and temporal properties are employed to train the PINN: Epoch data in raw format (T), Ω, n, M, e, ω, i, D_s, D_m, a and v. The periodic variation of D_s and D_m has resulted in the incorporation of these distances as training features for the LSTM model at each TLE epoch. RNNs frequently improve by incorporating additional pertinent features, as these features facilitate the models in comprehending intricate relationships and interactions among many aspects influencing satellite orbits. Significant enhancement in the model was observed with the utilization of these two elements (Mahmud et al., 2024).

For predicting each of the orbital elements, six different PINN models were employed as shown in Fig. 3. Each of the models has a unique customized loss function built with their respective governing equations discussed in Section 4.1. The PINN model is structured around an LSTM neural network optimized specifically for sequential prediction tasks. The LSTM network architecture is composed of interconnected gating mechanism, which enables the model to capture long-term dependencies and mitigate the vanishing gradient problem inherent in standard RNN. For all six models, the input data is segmented into fixed length sequences of 50-time steps, corresponding to most recent 50 epochs. This segmentation is important to capture the temporal patterns in the orbital data. The length of the sequence was empirically selected to balance both temporal coverage and computational efficiency, making sure that the network acquires adequate historical data to generate precise predictions without incurring significant memory expenses or gradient degradation.

A composite loss function designed to incorporate physics loss with data loss is used to train the model. The total loss is defined as Eq. (9), where, $\mathscr{L}_{\mathscr{D}}(\theta)$ denotes the data loss, which is computed as the MAE between the predicted orbital elements and the corresponding ground truth value. $\mathscr{L}_{\mathscr{F}}(\theta)$ denotes the physics loss, which introduces a regularization term that penalizes deviation from the known governing equations of orbital mechanics discussed in section 3. For computing the required derivatives of a function with time, AD was employed. The weights ω_F and ω_D are crucial in the performance of the network and need to be calibrated carefully to maintain an optimal trade-off between empirical accuracy and physical law enforcement.

To fine-tune the hyperparameters of the network, Bayesian Search algorithm was used to ensure robust learning. This algorithm dynamically models the objective function $x^* = \arg\max_{x\in X} f(x)$ by conducting a sequence of evaluations i.e. $x_1, \ldots, x_T$ of f, to identify the optimum of f in the most efficient way (Nguyen, 2019). The hyperparameters which are optimized include the number of LSTM units, layer, learning rate, batch size and dropout rate. The hyperparameter search space is presented in Table 1.

All models use the identical hyperparameter search space, but Bayesian optimization is performed separately for each orbital element to identify element-specific optimal configurations. This approach balances standardization across the framework with flexibility to accommodate the distinct prediction challenges of each orbital parameter.

As can be observed from Fig. 3, six models are trained for six orbital elements in TLE, where each of the modes receives the predicted value from the previous model as a test feature. This is why the sequence of training is important in this framework. Each PINN model is trained using a distinct combination of input features, uniquely tailored according to the specific orbital parameter being targeted. The precise combinations of the training datasets used for each model, along with the corresponding training sequence, are systematically presented in Table 2.

At the beginning of the training process, the immediately available training or input features are T, D_s and D_m. Thus, it is practical to begin with the model for n first, primarily because mean motion changes minimally over time, as clearly shown in Fig. 4. Additionally, a can be directly calculated from the predicted n, and this computed

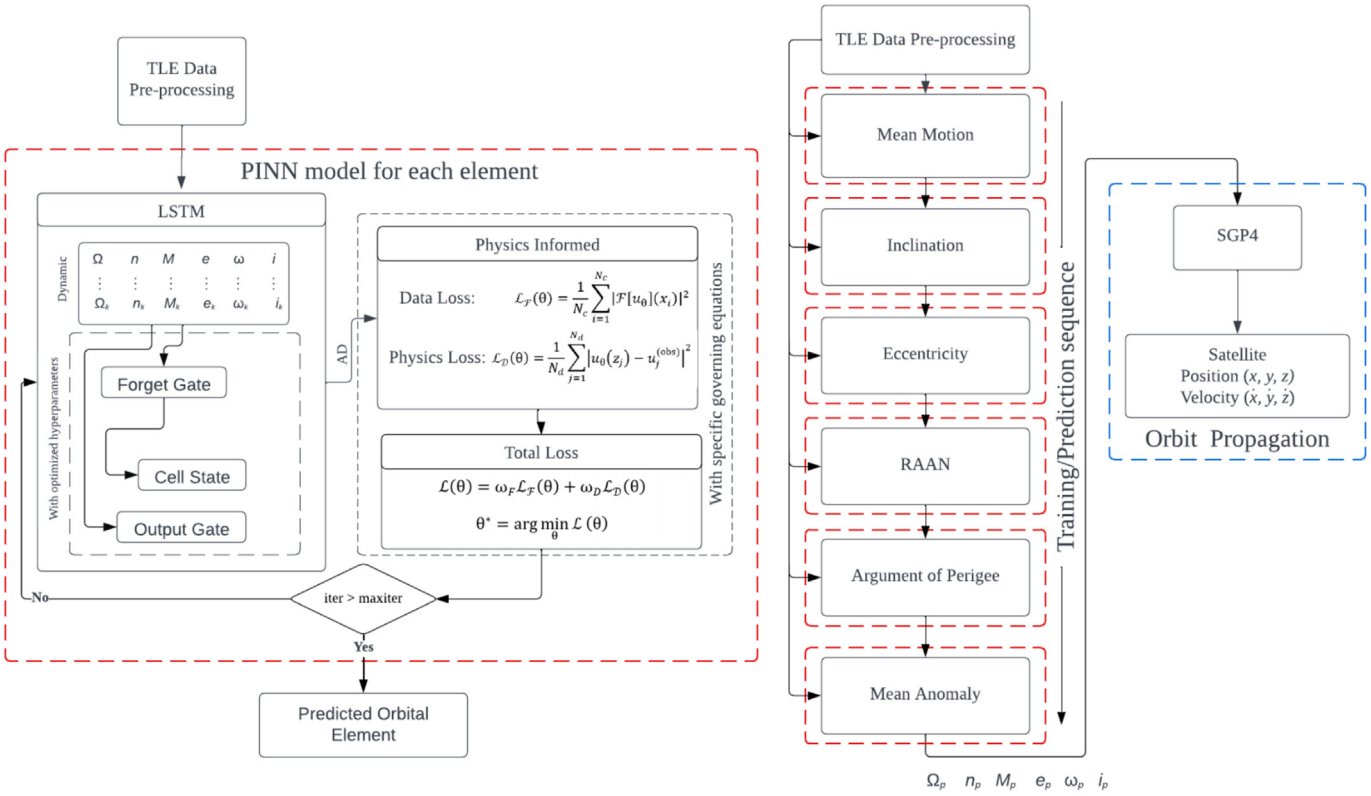


Fig. 3. Overview of the proposed PINN framework integrating LSTM for orbital parameter prediction. The model sequentially predicts each orbital element using TLE-derived features, incorporating both $\mathscr{L}_{\mathscr{F}}$ and $\mathscr{L}_{\mathscr{D}}$. The predicted elements are used with SGP4 for satellite state estimation.

Table 1
Search space for model hyperparameters.

Hyper-parameters	Search space
Unit	25–150
Dropout	0.1–0.5
Layer	1–9
Batch size	16–128
Learning rate	0.0001–0.01

Table 2
Unique combinations of training features for each of the six models (Mahmud et al., 2024).

Target element	Training features
Mean motion	T, D_s, D_m
Eccentricity	T, D_s, D_m, n, i, a
RAAN	T, D_s, D_m, n, i, e
Inclination	T, n
Mean Anomaly	T, D_s, D_m, n, i, a, v
Argument of Perigee	T, D_s, D_m, n, i, Ω

a becomes useful later when predicting e. Following the prediction of mean motion, the model for i is the next model to be trained as it also demonstrates limited variability, as illustrated by Fig. 4, making it straightforward and trustworthy to forecast early in the sequence with limited input features. e, exhibiting a clearly consistent pattern with minimal fluctuation (Fig. 4), it is thereafter predicted third, after the a has been calculated from the predicted n. e and a do not directly correlate; rather, collectively they define the orbit's general shape, so providing an extra critical feature for improved precision in the model. Ω is ranked fourth, making use of the aggregated parameters previously predicted (n, i and e), which improves the accuracy of the model. As can be observed from Fig. 4, ω has high variability in the historical data. Thus, it was placed at the fifth in the sequence. However, as the predicted ω was not used in any other model, placing it later will not affect the overall performance as well. For M, to obtain an acceptable prediction, it is important to construct the orbital plane first by predicting other parameters. Thus, the model for M is placed at the end of the sequence.

5. Experimental results

For experimental purposes, historical TLEs were collected from Celestrak, covering an extended period to capture diverse orbital conditions. IRIDIUM-7 and ORBCOMM FM26 were randomly selected, one with a comparatively large historical record (IRIDIUM 7 with 14,636 datapoints) and the other with a smaller dataset (ORBCOMM FM26 with 3802 datapoints). This was done to enable detailed, per-satellite evaluations and thus provide a focused assessment of the PINN model's accuracy on individual orbital trajectories. In addition, a separate set of seven Iridium satellites consisting of – IRIDIUM

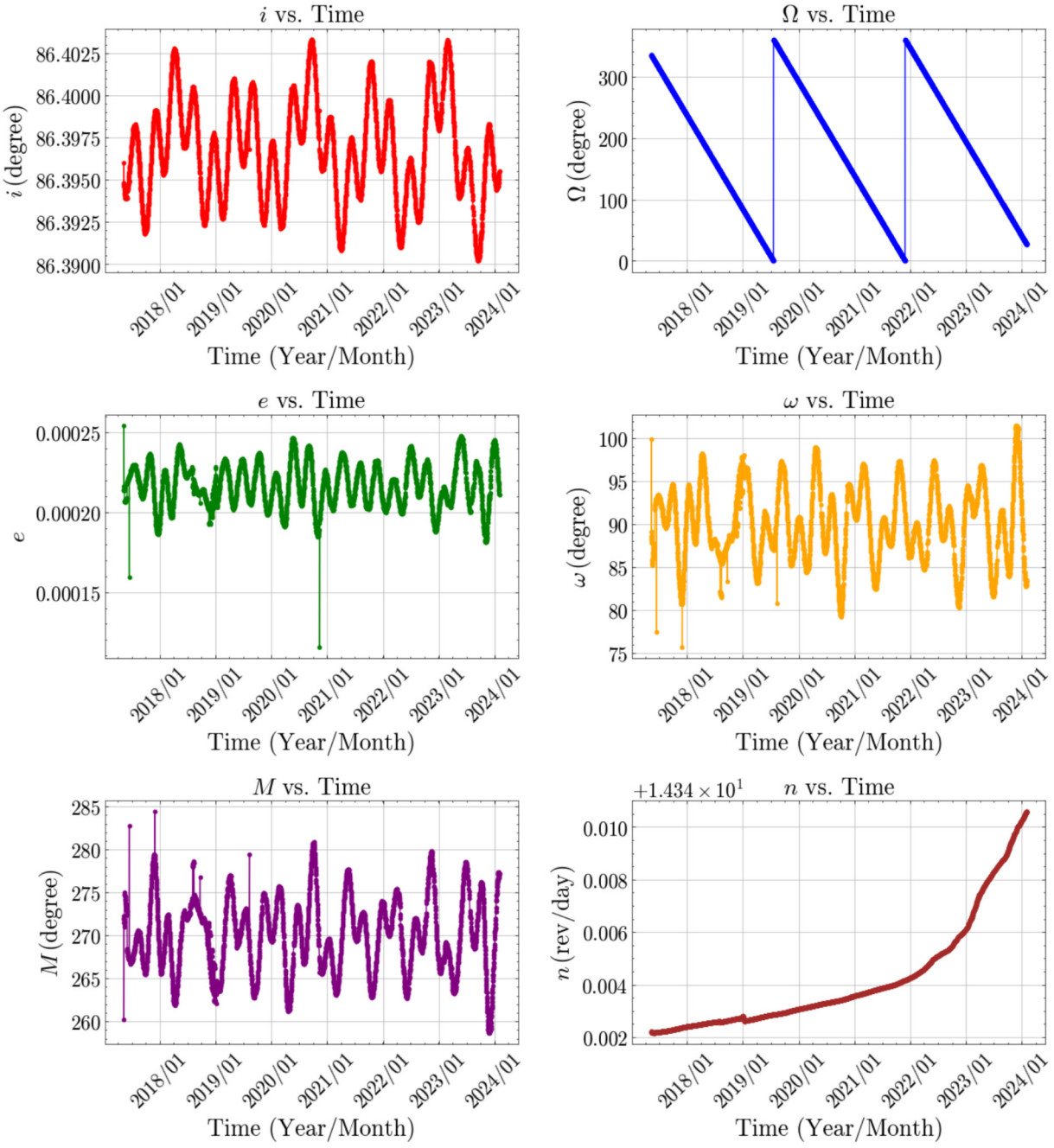


Fig. 4. Temporal evolution of orbital elements from the last 5000 IRIDIUM 7 TLE records.

16, IRIDIUM 17, IRIDIUM 26, IRIDIUM 28, IRIDIUM 38, IRIDIUM 39, and IRIDIUM 45 – was used to investigate the robustness of the models across multiple orbital parameters and to systematically study the influence of tuning physics loss weight. This two-stage experimental design provides both fine grained performance insights and a broader validation of the model's generalizability capabilities.

Experiments were first conducted on the 7 Iridium satellites to analyze the robustness of the models and study the importance of physics weight fine tuning. Plots in Fig. 5 show the percent MAE improvement over the baseline LSTM model (physics loss weight = 0) across a range of physics loss weight (0.1–0.9) for all six orbital elements.

The plots demonstrate that with an appropriate selection of the physics loss weight, the PINN consistently outperforms the baseline LSTM across all the satellites, confirming the model's robustness and general applicability. Although the weight varies across satellites and parameters, each satellite exhibits at least one weight value that yields a substantial accuracy gain, emphasizing flexibility and effectiveness of the models when the physics-informed regularization is tuned carefully. This indicates that the proposed model can be reliably deployed across

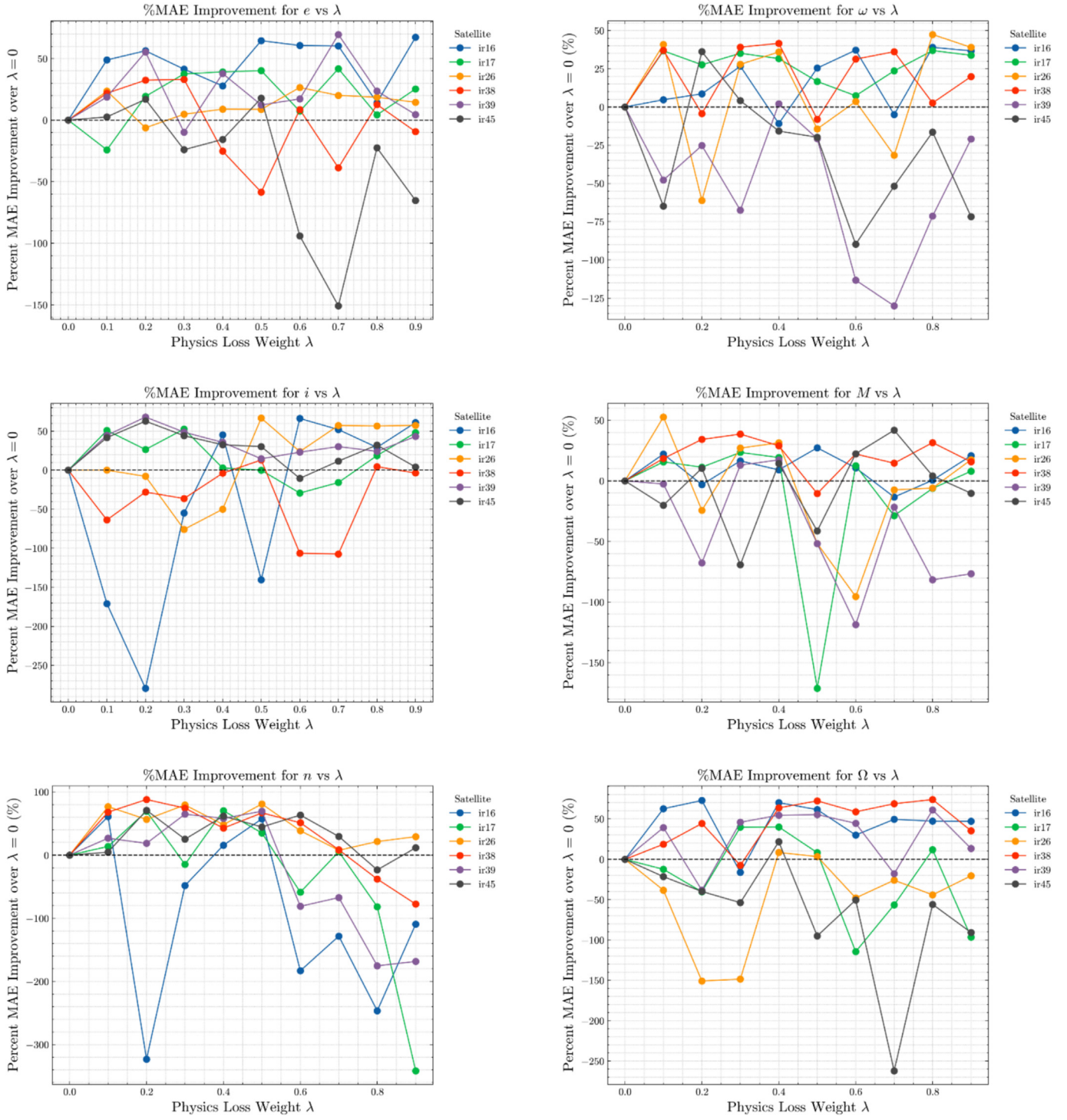


Fig. 5. Percent MAE improvement over the baseline LSTM for six orbital elements across seven Iridium satellites, illustrating the impact of physics-loss weight on model robustness and predictive accuracy.

a diverse set or orbital conditions, provided that the weight is adaptively optimized.

Experiments were then conducted with the dataset of IRIDIUM 7 to illustrate the critical role of physics-informed regularization, experiments were first conducted with varying physics loss weights. Fig. 6 depicts the predicted time-series for ω and i under different physics loss scenarios. The performance of a pure data-driven approach ($\mathscr{L}_{\mathscr{F}} = 0$) was compared against models with incremental physics constraints. Although the purely data-driven model first provides accurate short-term forecasts, it gradually deviates from the ground truth over a long period. Note that Fig. 6 presents selected $\mathscr{L}_{\mathscr{F}}$ values (0, 0.2, 0.4, 0.8) for visualization clarity. The optimal weights were

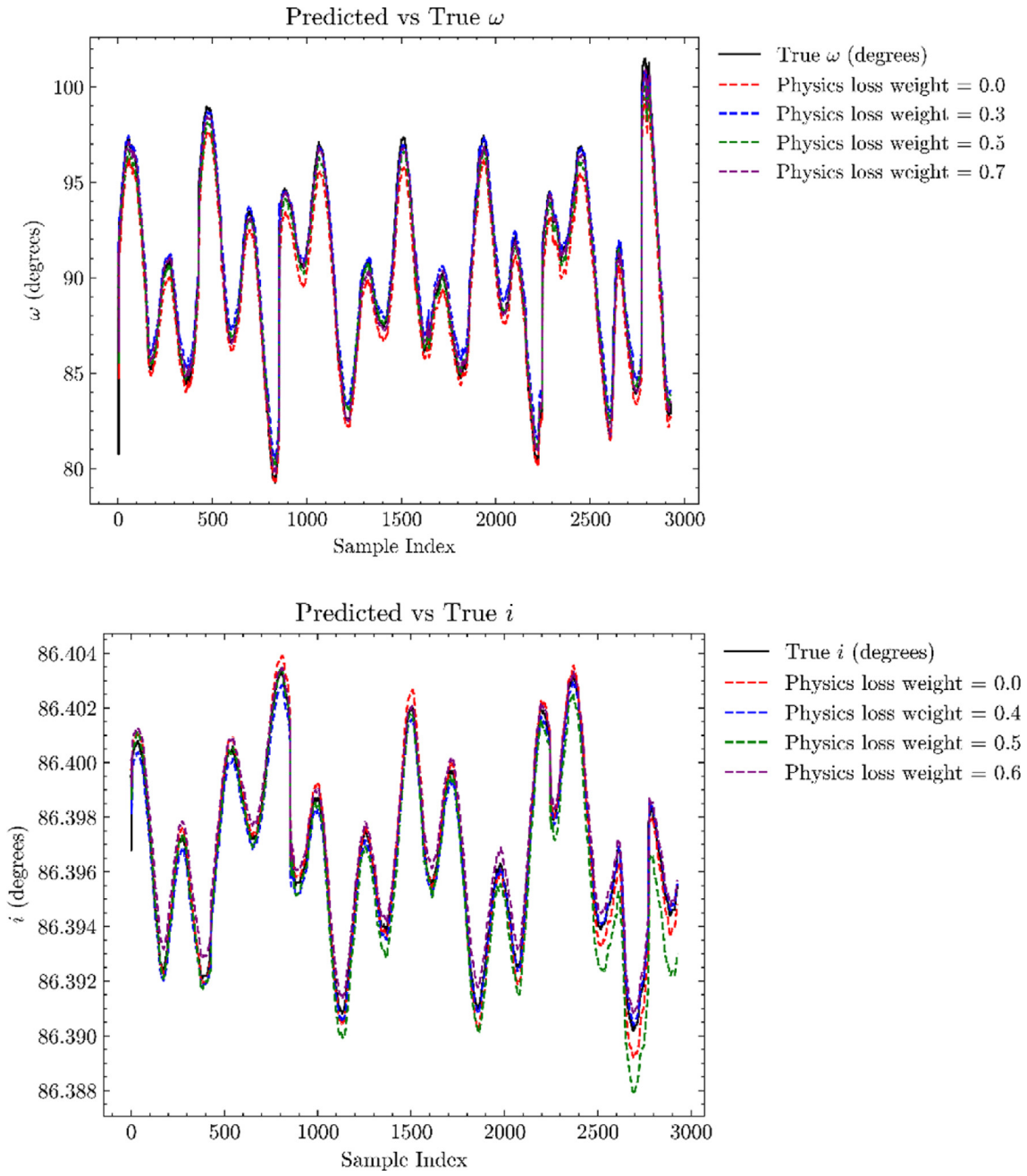


Fig. 6. Time-series comparison of the PINN predictions for ω and i under different physics loss weight settings for IRIDIUM 7.

determined through systematic manual tuning, evaluating weights from 0.1 to 0.9 in increments of 0.1. This comprehensive search, validated across seven satellites (Fig. 5), identified optimal weights for IRIDIUM 7. While this manual approach may seem labor-intensive, it was necessary given the different physical characteristics of each orbital parameter and ensures reproducibility of our results.

In the case of ω, the model trained with no physics constraints deviates from the known precession rates as errors accumulate over time. Incorporating a balanced physics loss constrains the model to adhere to the theoretical rate of change derived from the governing equations. For the model of i, similar behavior can be observed. The purely data-driven approach initially approximates the measured data well; however, as predictions extend further in time, the model starts to deviate from the ground truth. When a moderate physics loss is applied, especially when the weight is 0.4, the PINN predictions stabilize, with the model capturing both the oscillatory behavior and the long-term physical trend imposed by J_2 perturbation. It is crucial to note that simply incrementing physics loss weight beyond the optimal range does not guarantee enhanced performance; excessively high weights may impose severe

Table 3
RMSE, MAE, and R^2 score comparison of the base LSTM model and PINN for ω and i using optimum weights. Here, the coefficient of determination R^2 quantifies how much of the variability in the true values the model is able to explain. An R^2 score closer to 1 indicates a better fit to the data.

	ω (°)		i (°)	
	LSTM	PINN	LSTM	PINN
RMSE	0.0182	0.0131	0.00044	0.00035
MAE	0.0148	0.0093	0.00034	0.00028
R^2	0.9498	0.9738	0.9805	0.9875

constraints, thereby restricting the model's ability to accurately capture the empirical subtleties present in the data. This indicated that the proper tuning of the physics loss weight is crucial for achieving an optimal balance between data fidelity and physical consistency (Table 3).

Fig. 7 shows the comparison of base LSTM model predictions with PINN predictions for the other four elements – n, e, M and Ω. Plots indicate that for all the parameters, PINN outperforms LSTM, showing better generalization capabilities with physics-based corrections.

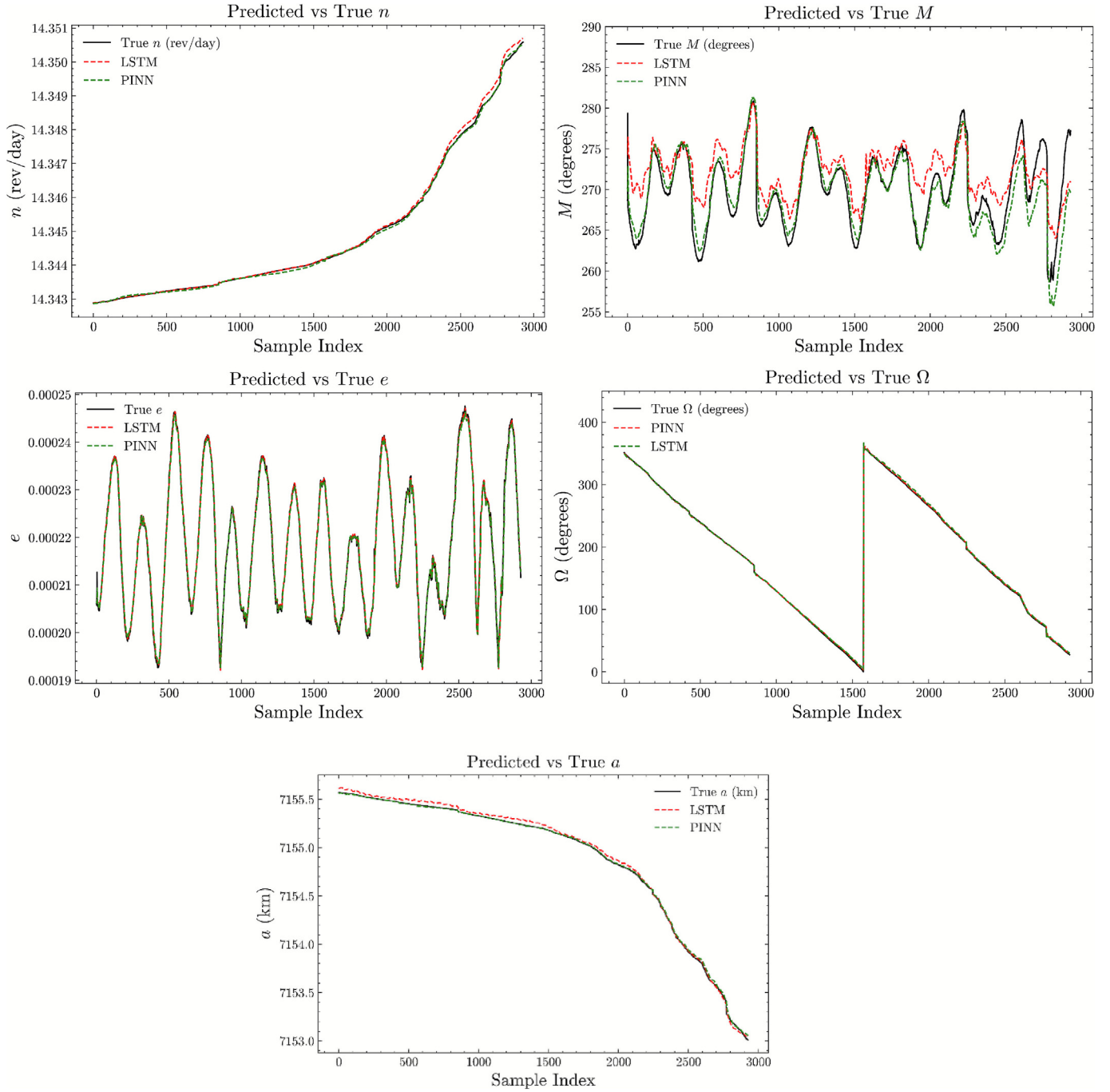


Fig. 7. Comparison of LSTM and PINN predictions against true orbital parameters: n, Ω, e, M and a for IRIDIUM 7. The plots illustrate the advantage of integrating physics-informed constraints within PINNs, particularly evident in improved long-term accuracy and stability.

In the case of n, both LSTM and PINN models exhibit a strong predictive accuracy in short term forecasting. However, PINN demonstrates superior long-term stability, which is the result of physics-based regularization that accounts for atmospheric drag induced orbital decay. For Ω, the predictions by both the models closely follow the secular drift associated with the earth's oblateness. PINN, however, shows greater resilience to abrupt transitions and long-term cumulative errors, reflecting its effective incorporation of physical constraints. In the case of e, both LSTM and PINN exhibit practically indistinguishable performance. This suggests that for parameters that are extremely periodic and exhibit gradual variation, such as e, the data-driven methodology alone adequately captures the fundamental dynamics, hence limiting the advantages of incorporating physics-informed constraints. The M plot clearly demonstrates the advantage of incorporating physics-informed loss functions. The purely data-driven LSTM exhibits significant deviation from the ground truth as predictions progress, primarily due to accumulating errors over time. In contrast, the PINN stabilizes these predictions by enforcing consistency with theoretical orbital dynamics, resulting in improved accuracy over extended prediction horizons. For a, as it was a deterministic term, improvement in n translates directly into corresponding improvements in a which can be observed in the figure.

Table 4
RMSE, MAE and R^2 score comparison of the base LSTM model and PINN for n, M, e and Ω using optimum weights.

	n (rev/day)		M (°)		e		Ω (°)		a (km)	
	LSTM	PINN	LSTM	PINN	LSTM	PINN	LSTM	PINN	LSTM	PINN
RMSE	8.41×10^{-5}	4.65×10^{-5}	0.0606	0.0442	8.47×10^{-7}	8.17×10^{-7}	0.1415	0.1175	0.0325	0.0132
MAE	4.84×10^{-5}	3.72×10^{-5}	0.0499	0.0373	6.36×10^{-7}	5.92×10^{-7}	0.0632	0.0210	0.0294	0.0084
R^2	0.9983	0.9995	0.4410	0.7018	0.9960	0.9966	0.8934	0.8954	0.9978	0.9996

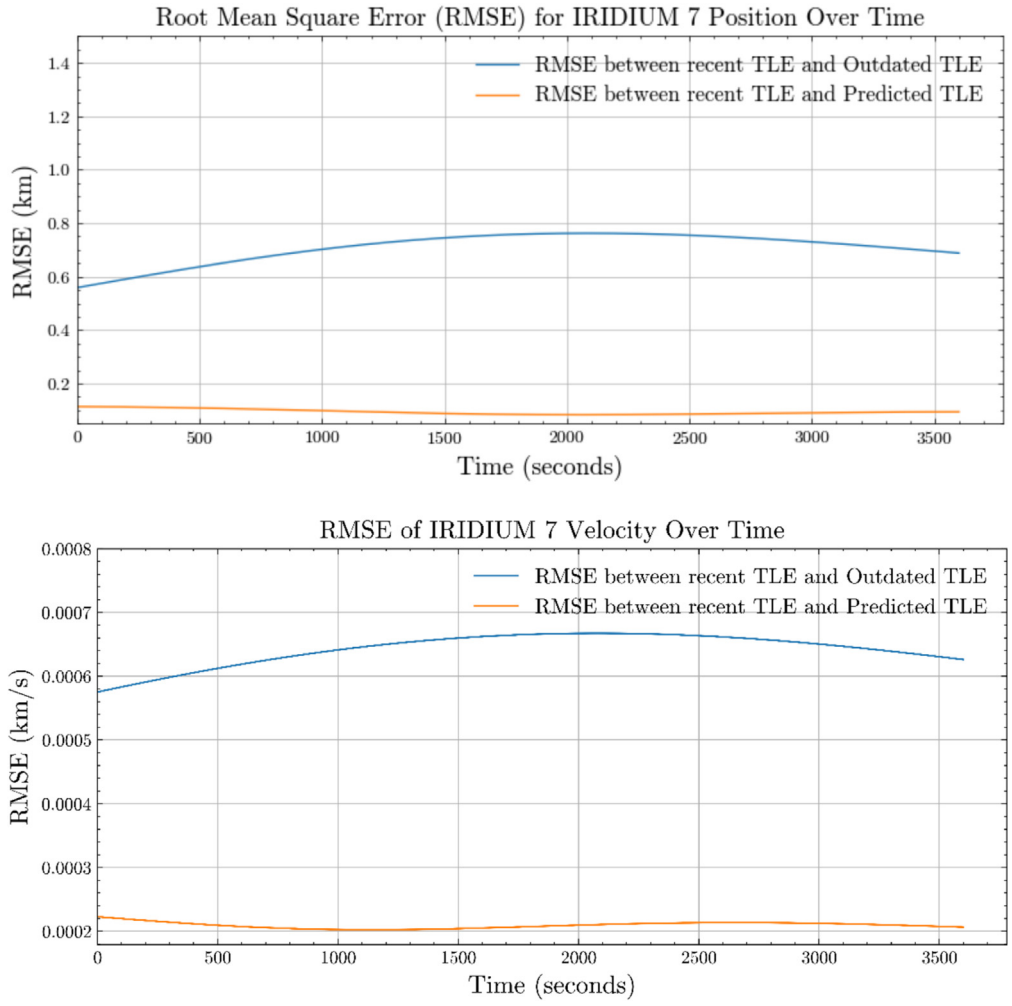


Fig. 8. RMSE for IRIDIUM 7 position and velocity over time, comparing the recent TLE with predicted TLE and an outdated TLE.

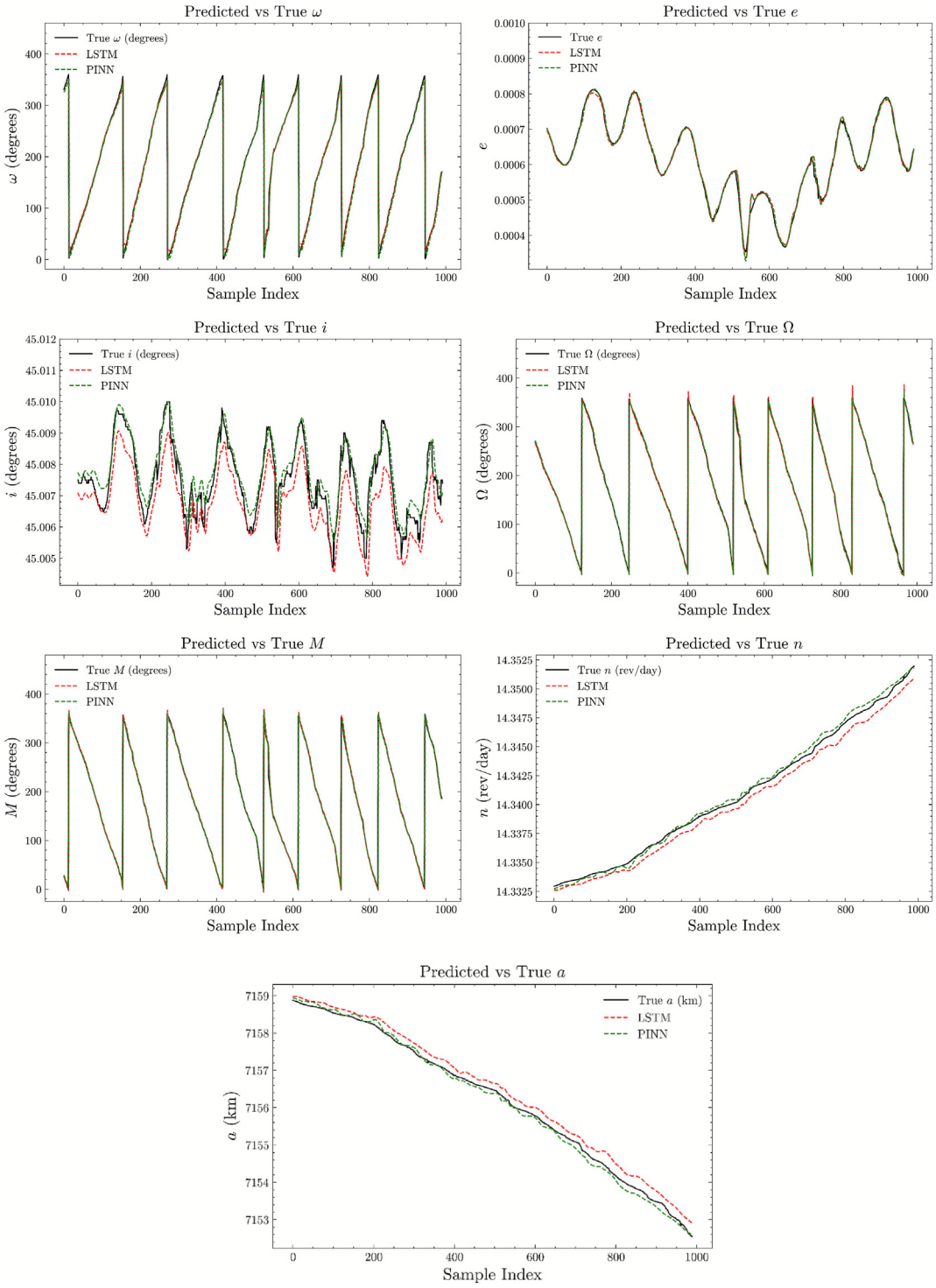


Fig. 9. Comparison showing LSTM predictions and PINN predictions for the orbital elements of ORBCOMM FM26 over 1000 epochs.

Table 5
RMSE and percentile error comparison for Ω predictions, highlighting performance near and away from the wrap-around region (0°/360°).

	RMSE (Near Wrap Region)		RMSE (Away Wrap Region)		P50		P95		P99	
	LSTM	PINN	LSTM	PINN	LSTM	PINN	LSTM	PINN	LSTM	PINN
Ω (°)	0.4415	0.3845	0.0777	0.0221	0.0443	0.0135	0.1852	0.0581	0.1985	0.1218

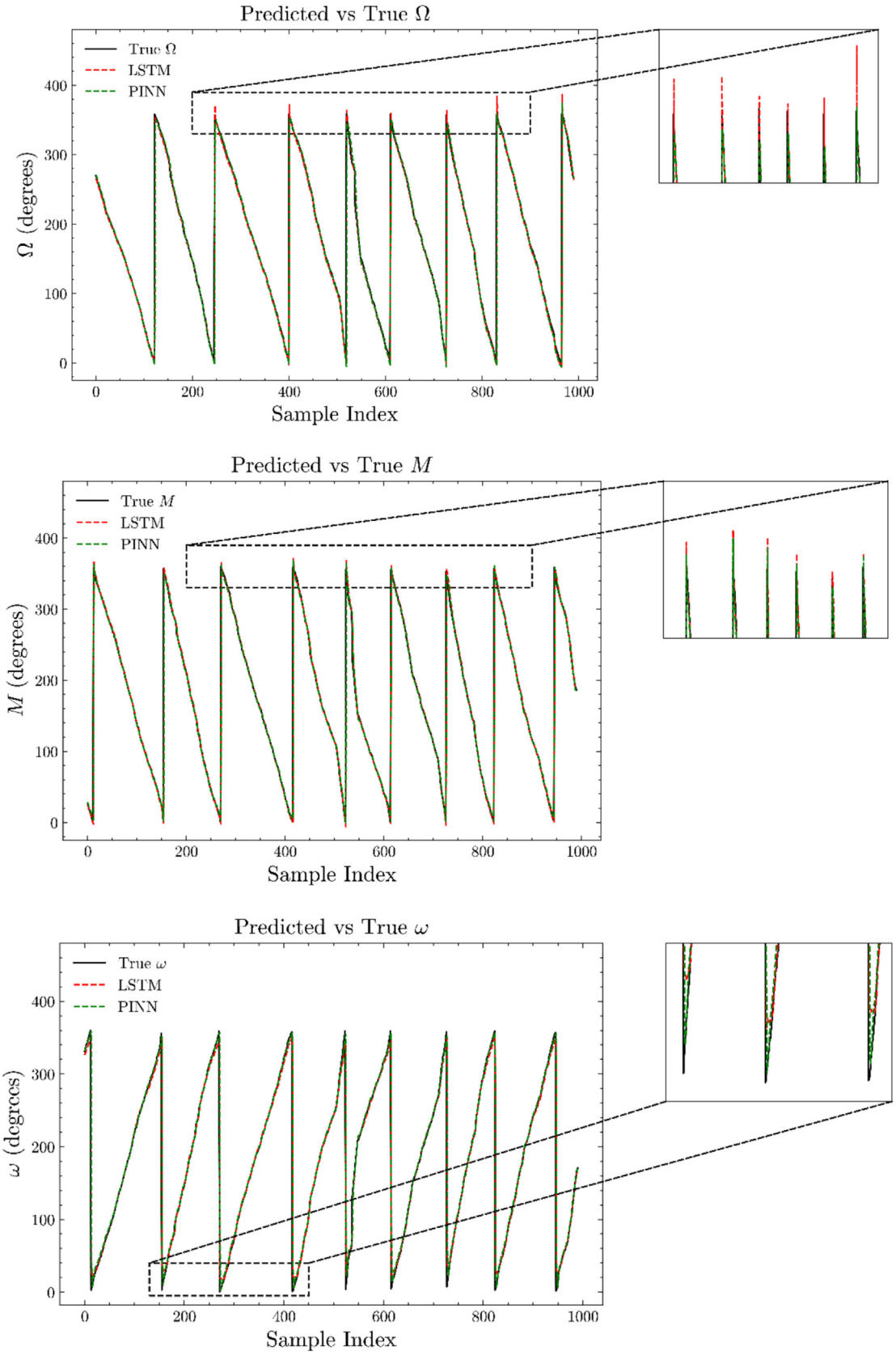


Fig. 10. Predicted vs. true Ω, M and ω for ORBCOMM FM26 using LSTM and PINN against the TLE reference. The inset zooms on the 0–360° wrap-around regions, where PINN delivers smoother, more continuous transitions and suppresses the abrupt discontinuities seen in the LSTM outputs.

Table 4 reflects the RMSE, MAE and R^2 score comparison of LSTM and PINN for each of the four models (Mahmud et al., 2024).

As can be seen from Table 4, the RMSE for Ω is comparatively higher than the RMSE of other parameters. This happens due to the inherent wrap-around discontinuity in angular measurements, where Ω resets from 360° back to 0°. These discontinuities are difficult to be captured by the model and even small deviations near this boundary can manifest as large absolute errors, which artificially inflate the RMSE. Although PINN reduces this error by some margin, it still affects the overall RMSE as can be seen from Table 5.

The model was then used to predict the six elements for the same epoch as the most recent TLE for IRIDIUM 7. The predicted values were then used with SGP4 propagator to propagate the satellite state for the next 3600 s. The most recent TLE and 25 h outdated TLE was also used to propagate the satellite states using SGP4. Fig. 8 shows the RMSE error between predicted and outdated TLEs with most TLE in position and velocity estimates.

The plots distinctly illustrate the superior predictive capability of the PINN approach, maintaining a low and nearly constant RMSE throughout the prediction horizon for both position and velocity estimates, peaking at around 100 m and 0.1 m/s respectively. However, the outdated TLE predictions exhibit significantly larger errors that increase over time, peaking at around 800 m and 0.7 m/s for position and velocity estimate respectively, indicating rapid degradation in predictive accuracy.

Similar experiments were then conducted on the dataset of ORBCOMM FM26, which had 3802 sets of TLE, 80% of which were used for training and rest for the testing phase. The purpose of using a different data set from a different constellation was to validate and understand the robustness and generalizability of the proposed PINN framework across different constellations – each characterized by unique operational conditions. Moreover, using a dataset with a comparatively smaller size provided insights into the model's predictive stability and reliability under conditions with different data availability. The plots in Fig. 9 show the proposed PINN's performance on the dataset from ORBCOMM FM26.

The plots in Fig. 9 illustrates that PINN performs better than baseline LSTM for all orbital parameter predictions. However, noticeable improvement is shown by the models of i and n, where the physics informed regularization strongly constrains prediction results (Table 5). For the case of i, the baseline LSTM shows a clear phase offset relative to the ground truth values of the parameter. This reflects the difficulty for the LSTM model to learn the subtle J_2 driven coupling between i and nodal precision from a limited dataset. However, embedding the physical forces through the governing equations, PINN remains phase locked to the ground truth even for a longer period.

A similar behavior is observed for the model of n, where the purely data-driven LSTM network replicates the long-term trend but accumulates a systematic bias that grows with time. However, incorporating physics into the loss function prevents the drift for PINN, yielding a visibly tighter fit compared to baseline LSTM predictions. This improvement for n directly translates into corresponding improvements for a as it is derived from the values of n.

For the remaining four parameters (ω, Ω, M, e), the predictions for solely data driven LSTM network shows a similar performance to the proposed PINN framework. The reason for such performance is the difference in satellite motion compared to IRIDIUM 7. For ORBCOMM FM26, ω, Ω, M and e changes gradually in the historical TLE with a clear pattern unlike IRIDIUM 7′s TLE dataset. However, PINN outperforms LSTM by damping high frequency artefacts and mitigating cycle-wrap discontinuities for ω, Ω and M. The purely data-driven LSTM network exhibits abrupt wrap discontinuities 0–360° boundaries, whereas PINN imposes continuity due to the embedded physics constraints (Fig. 10). Without compromising trend integrity, this regularization reduces spurious

Table 6
RMSE, MAE and R^2 score comparison of the base LSTM model and PINN for n, M, e, Ω, ω, i and a using optimum weights.

		RMSE	MAE	R^2
n (rev/day)	LSTM	0.0006	0.0006	0.9843
	PINN	0.0003	0.0002	0.9966
M (°)	LSTM	0.6007	0.0954	0.8901
	PINN	0.5931	0.0929	0.8929
Ω (°)	LSTM	0.5702	0.0960	0.8963
	PINN	0.4601	0.0923	0.8989
e	LSTM	8.29×10^{-6}	5.79×10^{-6}	0.9945
	PINN	7.71×10^{-6}	5.01×10^{-6}	0.9953
ω (°)	LSTM	0.5857	0.1377	0.8957
	PINN	0.5837	0.1121	0.8964
i (°)	LSTM	0.0008	0.0007	0.5278
	PINN	0.0004	0.0003	0.8418
a (km)	LSTM	0.2288	0.2196	0.9843
	PINN	0.0875	0.0827	0.9966

oscillations and ensures continuous phase advancement across cycle borders, hence producing cleaner, artifact-free predictions.

Table 6 shows that RMSE values for M, Ω and ω are comparatively higher than RMSE values for other parameters. As mentioned earlier, this occurs due to the inherent wrap-around discontinuity in angular parameters, where values reset from 360° back to 0°, causing small deviations near the boundary to appear as large errors and artificially inflating the metrics. While the incorporation of physics-informed constraints helps mitigate these effects by improving long-term stability, the wrap-around behavior still contributes to higher error levels compared to parameters with non-wrap-around behavior. Table 7 shows relevant statistical values that further decompose these errors, highlighting the impact of wrap regions and the extent to which PINN reduces extreme deviations.

The predicted values were then used with the SGP4 propagator to estimate the satellite state for the next 3600 s, similar to the prior experiment using the predicted values for the IRIDIUM 7 satellite. The latest TLE and a 24-h outdated TLE were used to calculate the satellite

Table 7
RMSE and percentile error comparison for M, Ω and ω predictions, highlighting performance near and away from the wrap-around region (0°/360°).

	RMSE (Near Wrap Region)		RMSE (Away Wrap Region)		P50		P95		P99	
	LSTM	PINN	LSTM	PINN	LSTM	PINN	LSTM	PINN	LSTM	PINN
M (°)	1.7926	1.3717	0.0688	0.0612	0.0254	0.0223	0.1277	0.1053	0.7355	0.5965
Ω (°)	1.7765	1.5451	0.0620	0.0603	0.0351	0.0339	0.1208	0.1101	0.5243	0.3652
ω (°)	1.7491	1.5424	0.1009	0.0685	0.0742	0.0342	0.2001	0.1878	0.6978	0.5884

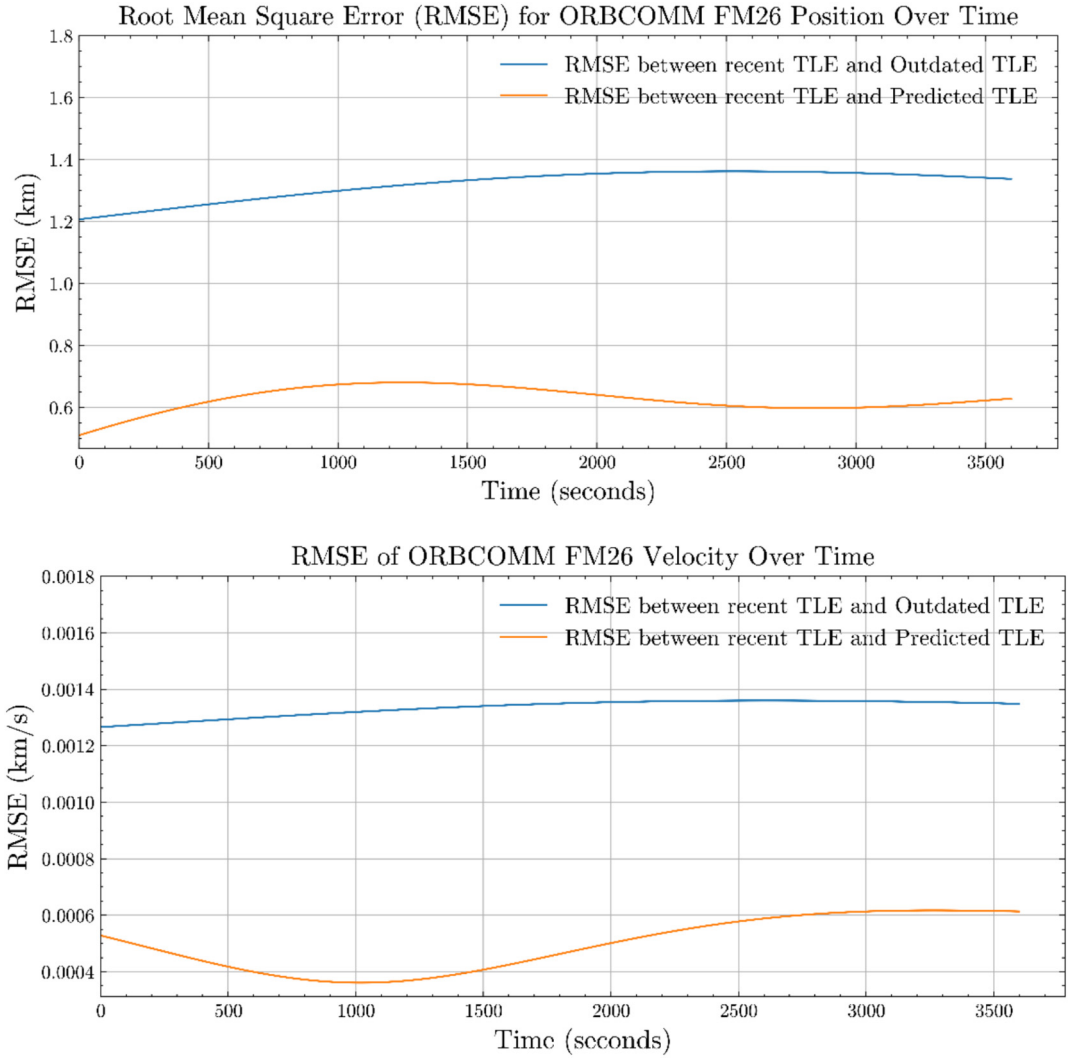


Fig. 11. RMSE for ORBCOMM FM26 position and velocity over time, comparing the recent TLE with predicted TLE and an outdated TLE.

states using SGP4. Fig. 11 shows the RMSE error between predicted and outdated TLEs with most TLE in position and velocity estimates.

Both the plots in Fig. 11 illustrate superior performance of PINN predicted orbital elements over an outdated TLE. This indicated the predicted TLE elements from the PINN model improve the accuracy in both position and velocity estimates. For ORBCOMM FM26 the position RMSE of the outdated TLE grows from about 1220 m to 1360 m over a one-hour horizon, whereas the PINN-predicted TLE stays below 700 m throughout—a nearly 50% improvement. Similarly, the velocity RMSE is improved almost 50%, from ~1.3 m/s down to ~0.6 m/s, with the PINN output exhibiting smaller oscillations and slower drift. This clear improvement demonstrates that even with a smaller training dataset, the physics-informed neural network yields robust, high-accuracy orbital forecasts—an asset for LEO positioning services and critical real-time space traffic management in increasingly crowded orbital regimes.

6. Conclusion and future work

This paper introduces a novel PINN based on LSTM framework that delivers quantitative improvements far beyond traditional TLE propagation. Specifically, for the IRIDIUM 7 satellite, the position and velocity RMSEs were reduced by 87% from ~650 m to ~85 m and 70% from ~0.65 m/s to ~0.2 m/s, respectively, while for ORBCOMM FM26 the RMSEs were reduced by 50% from ~1360 m to ~680 m and 72% from ~1.35 m/s to ~0.3 m/s, respectively. The proposed model has also demonstrated significant improvement over the baseline LSTM, achieving notable R^2 score enhancements, particularly for i, with an improvement of approximately 59.5%, and M, with an enhancement of approximately 59.1%. These improvements are achieved using only publicly available, sparsely updated TLE dataset, which clearly indicates that embedding relevant physical constraints into the network's loss function yields significant accuracy and more stable predictions. Furthermore, experiments conducted on seven additional Iridium satellites evaluated the robustness and generalization capabilities of the model. The percentage MAE improvements of each of the six orbital elements revealed that careful fine-tuning of the physics loss weighting consistently yielded substantial performance gains across diverse orbital conditions. Although optimal weight values varied slightly among satellites and orbital elements, these results underscore the critical importance and efficacy of physics weight selection. Moreover, the proposed framework is seen to be consistently outperforming a purely data-driven LSTM baseline model across all six orbital elements predictions – even when trained with a smaller dataset – by mitigating high-frequency noise and eliminating wrap-around discontinuities at 0-360° boundaries. Overall, these findings validate that the proposed PINN framework not only outperforms traditional propagation techniques but also ensures robust, accurate and reliable orbit prediction over multi orbit horizons.

Despite significant advancements, there exist several areas for future research. Primarily, future developments should aim at enhancing the current PINN framework for higher eccentricity LEO orbits by incorporating more comprehensive and higher-fidelity physical modeling, such as detailed atmospheric drag models (e.g., Jacchia-Roberts and NRLMSISE) and advanced gravitational field models extending beyond the J_2 perturbation approximation. Integrating additional perturbative effects, including solar radiation pressure and third-body gravitational influences, would likely improve model accuracy, particularly for extended forecasting horizons. Additionally, other deep learning models as base model and feature engineering can be explored to mitigate the problem of wrap-around discontinuity.

In conclusion, continued refinement and thorough validation of the proposed PINN framework are expected to significantly improve the operational reliability of satellite systems, leading to enhanced space sustainability and effective management of increasingly crowded orbital environments.

Declaration of Generative AI and AI-assisted technologies in the writing process

Author(s) have taken help of OpenAI's ChatGPT for Grammar corrections and formatting (LaTeX code generation for equations) during the preparation of this manuscript. After using the tool, the author(s) have reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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